Riemann–Roch Theorem via Holomorphic Line Bundles and the Picard Group on $\mathbb{CP}^1$

Ji, Yonghyeon

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We develop the Riemann–Roch theorem on $\mathbb{CP}^1$ through the following topics.

- The winding form $\frac{df}{f}$ as the universal counting tool
- Holomorphic line bundles $\mathcal{O}(n)$ and their classification
- Divisors and the associated line bundle $\mathcal{O}(D)$
- The Picard group $\mathrm{Pic}(\mathbb{CP}^1) \cong \mathbb{Z}$ via winding numbers
- The canonical bundle $K_{\mathbb{CP}^1} = \mathcal{O}(-2)$ and Serre duality
- Riemann–Roch on $\mathbb{CP}^1$ as a winding-number balance

Riemann–Roch Theorem. On a compact Riemann surface X of genus g,

$$\ell(D) - \ell(K - D) = \deg D + 1 - g,$$

where D is a divisor, K is a canonical divisor, and $\ell(D) = \dim_{\mathbb{C}} L(D)$. For $X = \mathbb{CP}^1$ one has $g = 0$, giving

$$\ell(D) - \ell(K - D) = \deg D + 1.$$

Guiding idea. The integers $\deg D$ and $\deg K$ are **winding numbers** — values of the integral $\frac{1}{2\pi i} \oint \frac{df}{f}$. Every quantity in the formula is assembled from this single primitive.

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Notation

$\mathcal{O}(X)$ Global holomorphic functions on X. **Warning:** $\mathcal{O}(X)$ uses a **manifold** argument and denotes a function ring; $\mathcal{O}(n)$ uses an **integer** argument and denotes a line bundle. These must never be confused.

$\mathcal{O}(n)$ Holomorphic line bundle on $\mathbb{CP}^1$ with transition function z^n

$\mathcal{M}(X)$ Meromorphic functions on X; $\mathcal{M}(X)^\times = \mathcal{M}(X) \setminus \{0\}$

$H^0(X, \mathcal{L})$ Space of global holomorphic sections of line bundle $\mathcal{L}$

$L(D)$ Riemann–Roch space $\{f \in \mathcal{M}(\mathbb{CP}^1)^\times : \operatorname{div}(f) + D \geq 0\} \cup \{0\}$

$\ell(D)$ $\dim_{\mathbb{C}} L(D)$

$\operatorname{Pic}(X)$ Picard group: isomorphism classes of holomorphic line bundles

$c_1(\mathcal{L})$ First Chern class = $\deg \mathcal{L}$ = winding number of transition function

$\frac{df}{f}$ **Winding form** of f: globally well-defined logarithmic derivative. Locally $\frac{df}{f} = d(\log f)$ wherever a branch of $\log f$ exists, but that equality is only a local mnemonic, not a global definition.

$\frac{1}{2\pi i}\oint_\gamma \frac{df}{f}$ Winding number of f along γ; equals $\operatorname{ord}_p(f)$ when γ is a small loop encircling only p

0 The Winding Form

Every integer appearing in this note — order of a zero or pole, degree of a divisor, winding number of a transition function — is the value of one integral. This section introduces that integral.

Coordinates on $\mathbb{CP}^1$

Complex Projective Line

Definition 1. $\mathbb{CP}^1 = (\mathbb{C}^2 \setminus \{0\})/\sim$ where $(Z_0, Z_1) \sim (\lambda Z_0, \lambda Z_1)$ for all $\lambda \in \mathbb{C}^\times$. The class is denoted $[Z_0 : Z_1]$.

Remark 1. The two standard affine charts:

$$U_0 = \{Z_0 \neq 0\}, \quad z = Z_1/Z_0, \qquad U_\infty = \{Z_1 \neq 0\}, \quad w = Z_0/Z_1.$$

On U_0: $[Z_0 : Z_1] = [1 : z]$. On U_∞: $[Z_0 : Z_1] = [w : 1]$. Overlap $U_0 \cap U_\infty \cong \mathbb{C}^\times$: $w = 1/z$. Point at infinity: $\infty = [0 : 1] \in U_\infty$ at $w = 0$.

0.1 The Logarithmic Derivative

Winding Form

Definition 2. For $f \in \mathcal{M}(U)^\times$, the **winding form** of f is the meromorphic 1-form

$$\frac{df}{f} = \frac{df}{f}.$$

Observation. On any open set where a holomorphic branch $\log f$ exists, $\frac{df}{f} = d(\log f)$ holds locally. However, $\frac{df}{f}$ is globally well-defined on all of U without branch choice: the ambiguity in $\log f$ (shifts by $2\pi i\mathbb{Z}$) vanishes upon differentiation. The expression "$d(\log f)$" is therefore only a local mnemonic.

Key computation. For $f = z^n$ ($n \in \mathbb{Z}$):

$$\frac{d(z^n)}{z^n} = \frac{nz^{n-1}\,dz}{z^n} = n\,\frac{dz}{z}.$$

Integrating around a small circle γ centred at 0:

$$\frac{1}{2\pi i}\oint_{\gamma}\frac{d\big((z^n)\big)}{(z^n)} = \frac{n}{2\pi i}\oint_{\gamma}\frac{dz}{z} = n.$$

This integer n is the order of f at 0 (positive for a zero, negative for a pole) and simultaneously the winding number of the map $z \mapsto z^n$ around the origin.

0.2 Winding Number = Order

Order via Winding

Definition 3. For $f \in \mathcal{M}(X)^{\times}$ and $p \in X$, the **order of** f **at** p is

$$\operatorname{ord}_p(f) := \frac{1}{2\pi i}\oint_{\gamma_p}\frac{d(f)}{f} = \frac{1}{2\pi i}\oint_{\gamma_p}\frac{df}{f},$$

where γ_p is a small loop encircling p once counterclockwise.

Observation. Write $f = z^m u$ near p, with local coordinate z centred at p and $u(p) \neq 0$. Then

$$\frac{df}{f} = m\,\frac{dz}{z} + \frac{du}{u}.$$

On a sufficiently small **simply connected** disk Δ around p where u is nonvanishing, u admits a holomorphic logarithm, so $\frac{du}{u} = d(\log u)$ is exact on Δ; hence $\oint_{\gamma_p}\frac{du}{u} = 0$. Only the pole of $\frac{dz}{z}$ contributes, giving $\operatorname{ord}_p(f) = m$.

0.3 The Global Stokes Argument: $\deg \operatorname{div}(f) = 0$

Lemma 1. *For $f \in \mathcal{M}(X)^{\times}$ on a compact Riemann surface X,*

$$\sum_{p\in X}\operatorname{ord}_p(f) = 0.$$

Proof. Let $p_1, \ldots, p_N$ be the zeros and poles of f. Choose pairwise disjoint closed disks $D_j(r)$ around each p_j with boundary circles $C_j(r) = \partial D_j(r)$, and set

$$X_r := X \setminus \bigcup_{j=1}^{N} D_j(r).$$

On X_r, the function f is holomorphic and nonvanishing, so locally $f = e^g$ for a holomorphic branch g. Thus $\omega := \dfrac{df}{f} = dg$ is locally exact, and

$$d\omega = d(dg) = 0$$

since $d \circ d = 0$. By Stokes:

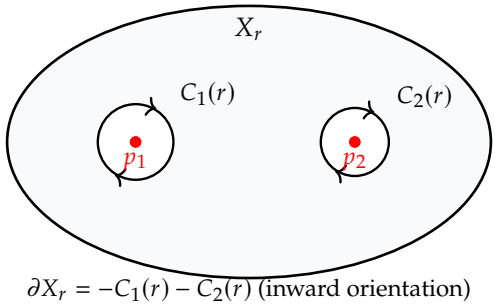
$$0 = \int_{X_r} d\omega = \int_{\partial X_r} \frac{df}{f}.$$

Since X has no boundary, the oriented boundary is $\partial X_r = -\sum_j C_j(r)$: removing each disk reverses the outward normal, so each $C_j(r)$ appears with the inward orientation. Therefore

$$0 = \int_{\partial X_r} \frac{df}{f} = -\sum_{j=1}^{N} \oint_{C_j(r)} \frac{df}{f} = -2\pi i \sum_{j=1}^{N} \operatorname{ord}_{p_j}(f), \tag{1}$$

giving $\displaystyle\sum_{j=1}^{N} \operatorname{ord}_{p_j}(f) = 0.$ □

The winding form $\dfrac{df}{f}$ is closed ($d\omega = 0$) wherever $f \neq 0, \infty$. Since $d\omega = 0$ on X_r, Stokes' theorem gives $\displaystyle\int_{\partial X_r} \omega = \int_{X_r} d\omega = 0$. It is this **boundary integral** that encodes all the winding numbers and must vanish — not an integral over the manifold itself.



$\partial X_r = -C_1(r) - C_2(r)$ (inward orientation)

$$0 = \int_{X_r} d\omega = \int_{\partial X_r} \omega = -\sum_j \oint_{C_j(r)} \frac{df}{f} = -2\pi i \sum_j \operatorname{ord}_{p_j}(f)$$

0.4 An Explicit Example on $\mathbb{CP}^1$

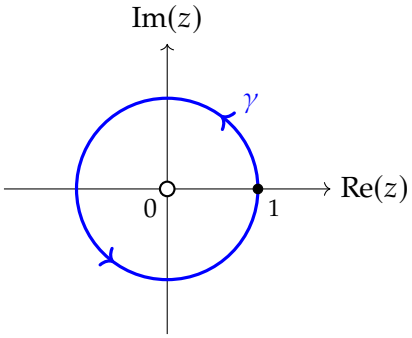
Example 1 (Winding form of $z - a$). Let $f = z - a$ for $a \in \mathbb{C}$. In the z-chart:

$$\frac{d(z-a)}{z-a} = \frac{dz}{z-a}, \qquad \operatorname{ord}_a(f) = \frac{1}{2\pi i} \oint_{\gamma_a} \frac{dz}{z-a} = 1.$$

Near ∞, pass to $w = 1/z$:

$$\left.\frac{d(z-a)}{z-a}\right|_{w=1/z} = \frac{-w^{-2}\,dw}{w^{-1}-a} = \frac{-w^{-1}\,dw}{1-aw}, \qquad \operatorname{ord}_\infty(f) = -1.$$

Total: $1 + (-1) = 0$. ✓ This verifies Lemma 1 from the winding form directly.



$$\frac{1}{2\pi i}\oint_\gamma \frac{dz}{z} = 1$$

1 Holomorphic Line Bundles on $\mathbb{CP}^1$

Every integer-valued invariant of a line bundle on $\mathbb{CP}^1$ is a winding number.

1.1 Definition via Transition Functions

Holomorphic Line Bundle

Definition 4. A **holomorphic line bundle** $\mathcal{L} \to X$ is a rank-1 complex vector bundle with holomorphic transition functions $g_{ij} : U_i \cap U_j \to \mathbb{C}^\times$ satisfying the cocycle condition $g_{ij}g_{jk} = g_{ik}$ on every triple overlap.

Remark 2. On $\mathbb{CP}^1 = U_0 \cup U_\infty$, the overlap $U_0 \cap U_\infty \cong \mathbb{C}^\times$ has $\pi_1(\mathbb{C}^\times) \cong \mathbb{Z}$, generated by the unit circle $|z| = 1$. A line bundle is determined (up to isomorphism) by a single transition function $g_{0\infty} : \mathbb{C}^\times \to \mathbb{C}^\times$, whose homotopy class is its winding number:

$$n = \frac{1}{2\pi i} \oint_{|z|=1} \frac{d(g_{0\infty})}{g_{0\infty}}.$$

Two transition functions with the same winding number give isomorphic bundles *(proved in Section 3)*. This is why line bundles on $\mathbb{CP}^1$ are classified by $\mathbb{Z}$.

1.2 The Line Bundles $\mathcal{O}(n)$

$\mathcal{O}(n)$

Definition 5. For $n \in \mathbb{Z}$, the line bundle $\mathcal{O}(n)$ on $\mathbb{CP}^1$ has transition function

$$g_{0\infty}(z) = z^n.$$

Special cases:

- $\mathcal{O}(0)$: trivial bundle ($g = 1$, winding number 0).
- $\mathcal{O}(1)$: hyperplane bundle ($g = z$, winding number 1).
- $\mathcal{O}(-1)$: tautological bundle ($g = z^{-1}$, winding number -1).

Example 2 (Winding number of $\mathcal{O}(n)$)**.** From §0.1:

$$\frac{1}{2\pi i} \oint_{|z|=1} \frac{d((z^n))}{(z^n)} = n.$$

The winding number of the transition function equals the label n. ✓

Observation. Two transition functions $g = z^n$ and $g' = e^{i\theta}z^n$ give isomorphic bundles because they differ by a coboundary: set $\phi_0 \equiv 1$ on U_0 and $\phi_\infty \equiv e^{i\theta}$ on U_∞. Then $g'_{0\infty} = \phi_0^{-1} g_{0\infty} \phi_\infty$. Both ϕ_i are holomorphic and nonvanishing on their simply connected charts, so $\phi_i = e^{h_i}$ for entire functions h_i. The coboundary contributes winding number

$$\frac{1}{2\pi i} \oint_{|z|=1} d(h_0 - h_\infty) = 0$$

(integral of an exact form on a closed loop). Hence only the winding number of $g_{0\infty}$ survives as an invariant.

1.3 Global Sections: A Winding Constraint

Proposition 2. *For $n \geq 0$,*

$$H^0\big(\mathbb{CP}^1, \mathcal{O}(n)\big) = \operatorname{span}_{\mathbb{C}}\{1, z, z^2, \ldots, z^n\}, \qquad h^0\big(\mathcal{O}(n)\big) = n + 1.$$

For $n < 0$, $H^0\big(\mathbb{CP}^1, \mathcal{O}(n)\big) = 0$.

Proof. A global section is a pair (s_0, s_∞) of holomorphic functions on U_0 and U_∞ satisfying the gluing condition

$$s_0(z) = z^n \cdot s_\infty\left(\tfrac{1}{z}\right) \quad \text{for all } z \in U_0 \cap U_\infty.$$

Step 1. Set $s_0(z) = P(z)$, a holomorphic (hence polynomial) function on $U_0 \cong \mathbb{C}$. Rearranging the gluing:

$$s_\infty(w) = w^n \, P\big(\tfrac{1}{w}\big).$$

Derivation: from $s_0(z) = z^n \cdot s_\infty(1/z)$ set $w = 1/z$ to get $s_\infty(w) = w^n s_0(1/w) = w^n P(1/w)$.

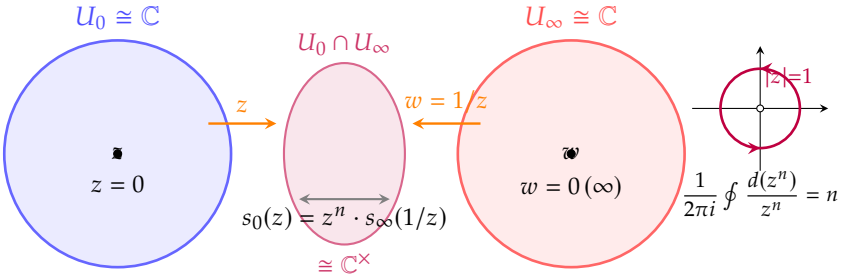
Step 2. s_∞ must be holomorphic at $w = 0$ (the point ∞). Write $P(z) = \sum_{k=0}^{m} a_k z^k$. Then

$$s_\infty(w) = \sum_{k=0}^{m} a_k \, w^{n-k}.$$

Holomorphicity at $w = 0$ requires $n - k \geq 0$ for every $k \leq m$, i.e., $\deg P \leq n$.

Conclusion. If $n \geq 0$: P is any polynomial of degree $\leq n$, giving $\operatorname{span}_{\mathbb{C}}\{1, \ldots, z^n\}$ of dimension $n + 1$. If $n < 0$: $\deg P \leq n < 0$ forces $P \equiv 0$, so $H^0 = 0$. □

The gluing $s_0(z) = z^n \cdot s_\infty(1/z)$ is the winding constraint: the section must **twist** n **times** as it crosses the equator $|z| = 1$. For $n \geq 0$ there are $n+1$ independent ways to do this; for $n < 0$ none.



gluing: $w = 1/z$

2 From Divisors to Line Bundles

A divisor encodes winding data at each point. This section assembles those data into transition functions and shows that $H^0(\mathcal{O}(D)) = L(D)$.

2.1 The Line Bundle $\mathcal{O}(D)$

Line Bundle Associated to a Divisor

Definition 6. Let $D = \sum_p n_p[p] \in \mathrm{Div}(\mathbb{CP}^1)$.

(a) **Sheaf of sections.** $\mathcal{O}(D)$ assigns to each open U the set

$$\mathcal{O}(D)(U) := \left\{ f \in \mathcal{M}(U)^\times : \mathrm{div}(f)\big|_U + D\big|_U \geq 0 \right\} \cup \{0\}.$$

(b) **Transition functions (equivalent description).** Choose $\{U_i\}$ fine enough that each U_i contains at most one support point of D. On each U_i pick $\psi_i \in \mathcal{M}(U_i)^\times$ with $\mathrm{div}(\psi_i) = D|_{U_i}$. Set

$$g_{ij} := \frac{\psi_i}{\psi_j} \quad \text{on } U_i \cap U_j.$$

Then g_{ij} is holomorphic and nonvanishing (since $\mathrm{div}(g_{ij}) = D|_{U_i} - D|_{U_j} = 0$ on the overlap) and satisfies the cocycle condition.

Remark 3 (Equivalence of the two descriptions)**.** A section s of the sheaf over U_i satisfies $\mathrm{div}(s)|_{U_i} + D|_{U_i} \geq 0$, i.e., $s = \lambda_i \psi_i$ for a holomorphic function λ_i. Thus λ_i is the fiber coordinate in the ψ_i-trivialization, and the transition $\lambda_i = g_{ij}\lambda_j$ on overlaps recovers the line bundle structure. The two descriptions define the same object.

Remark 4 (Well-definedness)**.** Replacing ψ_i by $f_i\psi_i$ (with f_i holomorphic nonvanishing on U_i) gives $g'_{ij} = f_i g_{ij} f_j^{-1}$, a coboundary with $\phi_k = f_k^{-1}$. The resulting bundle is isomorphic to the original, so $\mathcal{O}(D)$ is well-defined as an isomorphism class.

Observation (Winding data)**.** The winding number of ψ_i around a single point $p \in U_i$ is

$$\frac{1}{2\pi i} \oint_{\gamma_p} \frac{d\psi_i}{\psi_i} = \mathrm{ord}_p(\psi_i) = n_p.$$

The transition functions $g_{ij} = \psi_i/\psi_j$ are nonvanishing on overlaps, but the local trivializations ψ_i carry the winding data of D pointwise.

Concrete case: $D = n[\infty]$. Take $U_0 \ni \mathbb{C}$ (which does not contain ∞): set $\psi_0 = 1$. Take U_∞ with ∞ at $w = 0$: set $\psi_\infty = w^n$. On the overlap, with $z = 1/w$:

$$g_{0\infty}(z) = \frac{\psi_0(z)}{\psi_\infty(1/z)} = \frac{1}{(1/z)^n} = z^n.$$

Winding number n, so $\mathcal{O}(n[\infty]) = \mathcal{O}(n)$. ✓

Theorem 3. *The map $D \mapsto \mathcal{O}(D)$ is a group homomorphism* $\mathrm{Div}(\mathbb{CP}^1) \to \mathrm{Pic}(\mathbb{CP}^1)$.

Proof. Adding divisors corresponds to multiplying local trivializations: $\psi_i^{(D+D')} = \psi_i^{(D)} \cdot \psi_i^{(D')}$, and therefore $g_{ij}^{(D+D')} = g_{ij}^{(D)} \cdot g_{ij}^{(D')}$. Multiplying transition functions corresponds to tensoring bundles: $\mathcal{O}(D + D') = \mathcal{O}(D) \otimes \mathcal{O}(D')$. □

2.2 Global Sections Recover $L(D)$

Theorem 4. $H^0\big(\mathbb{CP}^1, \mathcal{O}(D)\big) = L(D)$.

Proof. By Definition 6(a), a global section of $\mathcal{O}(D)$ is exactly a global meromorphic function $f \in \mathcal{M}(\mathbb{CP}^1)$ satisfying $\mathrm{div}(f) + D \geq 0$ (together with the zero function). This is the definition of $L(D)$. □

The winding form perspective makes $H^0(\mathcal{O}(D)) = L(D)$ transparent: both encode the constraint that the winding number of f at each point p is at least $-n_p$.

2.3 Linear Equivalence and Isomorphism

Theorem 5. $D \sim D' \iff \mathcal{O}(D) \cong \mathcal{O}(D')$.

Proof. **(⇒)** Suppose $D - D' = \mathrm{div}(g)$ for some $g \in \mathcal{M}(\mathbb{CP}^1)^\times$. Take local trivializations ψ_i for D. Define $\psi_i' := \psi_i / g$; then $\mathrm{div}(\psi_i') = D|_{U_i} - \mathrm{div}(g)|_{U_i} = D'|_{U_i}$. The transition functions:

$$g_{ij}' = \frac{\psi_i'}{\psi_j'} = \frac{\psi_i / g}{\psi_j / g} = \frac{\psi_i}{\psi_j} = g_{ij}.$$

Identical transition functions give isomorphic bundles.

(⇐) If $\mathcal{O}(D) \cong \mathcal{O}(D')$, the isomorphism supplies holomorphic nonvanishing functions ϕ_i on each U_i with $g'_{ij} = \phi_i^{-1} g_{ij} \phi_j$. Define

$$h_i := \frac{\phi_i^{-1}\psi_i}{\psi'_i} \quad \text{on } U_i.$$

On each overlap $U_i \cap U_j$, using $g_{ij}/g'_{ij} = \phi_i/\phi_j$ (from $g'_{ij} = \phi_i^{-1} g_{ij} \phi_j$):

$$\frac{h_i}{h_j} = \frac{\phi_j}{\phi_i} \cdot \frac{\psi_i/\psi'_i}{\psi_j/\psi'_j} = \frac{\phi_j}{\phi_i} \cdot \frac{g_{ij}}{g'_{ij}} = \frac{\phi_j}{\phi_i} \cdot \frac{\phi_i}{\phi_j} = 1.$$

So h_i patches to a global meromorphic h. Its divisor: $\operatorname{div}(h_i) = \operatorname{div}(\phi_i^{-1}) + \operatorname{div}(\psi_i) - \operatorname{div}(\psi'_i) = 0 + D|_{U_i} - D'|_{U_i} = (D - D')|_{U_i}$, giving $\operatorname{div}(h) = D - D'$, hence $D \sim D'$.

Note: patching ψ_i/ψ'_i alone fails because $(\psi_i/\psi'_i)/(\psi_j/\psi'_j) = g_{ij}/g'_{ij} = \phi_i \phi_j^{-1} \neq 1$ in general; the coboundary ϕ_i from the isomorphism is essential. □

Corollary 6. *If $D \sim D'$ then $\ell(D) = \ell(D')$.*

Proof. The isomorphism $\mathcal{O}(D) \cong \mathcal{O}(D')$ induces $H^0(\mathcal{O}(D)) \cong H^0(\mathcal{O}(D'))$, hence $\ell(D) = \ell(D')$ by Theorem 4. For an explicit vector-space isomorphism: if $\operatorname{div}(g) = D - D'$, then

$$\begin{aligned} \Phi_g : L(D) &\longrightarrow L(D') \\ f &\longmapsto gf \end{aligned}$$

is an isomorphism (see [1] for details). □

3 The Picard Group $\mathrm{Pic}(\mathbb{CP}^1)$

The degree of a line bundle is the winding number of its transition function around the equator. On $\mathbb{CP}^1$ this single integer is the only invariant.

3.1 Definition

Picard Group

Definition 7. $\mathrm{Pic}(X)$ is the group of isomorphism classes of holomorphic line bundles on X, with group operation $\otimes$ (tensor product) and identity $\mathcal{O}_X$. The inverse of $[\mathcal{L}]$ is $[\mathcal{L}^{-1}] = [\mathcal{L}^\vee]$.

3.2 The Degree Map as a Winding Number

Degree of a Line Bundle

Definition 8. The **degree** of $\mathcal{L}$ with transition function $g_{0\infty}$ is

$$\deg \mathcal{L} := \frac{1}{2\pi i}\oint_{|z|=1} \frac{d(g_{0\infty})}{g_{0\infty}} = \frac{1}{2\pi i}\oint_{|z|=1} \frac{dg_{0\infty}}{g_{0\infty}}.$$

Remark 5 (Well-definedness). Two representatives of $[\mathcal{L}]$ differ by a coboundary $g'_{0\infty} = \phi_0^{-1} g_{0\infty}\phi_\infty$, where ϕ_0, ϕ_∞ are holomorphic nonvanishing on $U_0 \cong \mathbb{C}$ and $U_\infty \cong \mathbb{C}$ respectively. Both charts are simply connected, so $\phi_i = e^{h_i}$ for entire h_i, and the coboundary contributes

$$\frac{1}{2\pi i}\oint_{|z|=1} d(h_\infty - h_0) = 0.$$

Hence $\deg \mathcal{L}$ is independent of the representative.

Important: the coboundary $\phi_0^{-1}\phi_\infty$ is defined only on the overlap $U_0 \cap U_\infty \cong \mathbb{C}^\times$, not on all of $\mathbb{CP}^1$. Lemma 1 (the degree-zero law) applies to globally defined functions and cannot be used here; simple connectivity of the charts is the correct tool.

Remark 6 ($\deg \mathcal{O}(D) = \deg D$). Assume D has no support on $|z| = 1$ (generic). With $g_{0\infty} = \psi_0/\psi_\infty$ and $d(gg')/(gg') = dg/g + dg'/g'$:

$$\frac{1}{2\pi i}\oint_{|z|=1} \frac{dg_{0\infty}}{g_{0\infty}} = \underbrace{\frac{1}{2\pi i}\oint_{|z|=1} \frac{d\psi_0}{\psi_0}}_{=\sum_{|p|<1} n_p} - \underbrace{\frac{1}{2\pi i}\oint_{|z|=1} \frac{d\psi_\infty}{\psi_\infty}}_{=-\sum_{|p|>1 \text{ or } p=\infty} n_p} = \sum_p n_p = \deg D.$$

The **first** term is the argument principle (ccw in z) counting zeros/poles of ψ_0 inside $|z| < 1$. The

minus sign in the second comes from orientation reversal: $|z| = 1$ counterclockwise in z becomes $|w| = 1$ **clockwise** in $w = 1/z$ (same mechanism as $\partial X_r = -\sum_j C_j(r)$ in §0.3); the argument principle in the w-chart then gives $-\sum_{|p|>1 \text{ or } p=\infty} n_p$.

Proposition 7. $\deg : \mathrm{Pic}(\mathbb{CP}^1) \to \mathbb{Z}$ *is a group homomorphism.*

Proof. The tensor product $\mathcal{L} \otimes \mathcal{L}'$ has transition function gg', and

$$\frac{d(gg')}{gg'} = \frac{dg}{g} + \frac{dg'}{g'},$$

so $\deg(\mathcal{L} \otimes \mathcal{L}') = \deg \mathcal{L} + \deg \mathcal{L}'$. □

3.3 $\mathrm{Pic}(\mathbb{CP}^1) \cong \mathbb{Z}$

Theorem 8.

$$\Phi\colon \mathrm{Pic}(\mathbb{CP}^1) \longrightarrow \mathbb{Z}$$
$$[\mathcal{L}] \longmapsto \deg \mathcal{L}$$

is a group isomorphism.

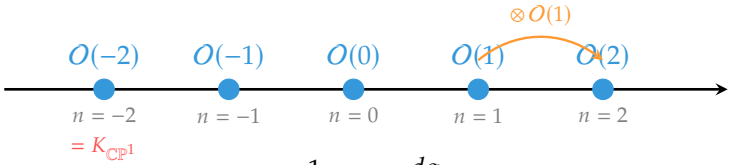
Proof. **Key lemma** (proved in [1]): **every divisor** D **on** $\mathbb{CP}^1$ **satisfies** $D \sim (\deg D)[\infty]$**.** The proof uses $\mathrm{div}(z - a) = [a] - [\infty]$ to move each finite point to $[\infty]$. This single lemma yields both surjectivity and injectivity without circularity.

Surjectivity: z^n achieves every integer winding number, so $\mathcal{O}(n)$ is a line bundle of degree n for every $n \in \mathbb{Z}$.

Injectivity: Suppose $\deg[\mathcal{L}] = n$. Any line bundle $\mathcal{L}$ of degree n has a meromorphic section s (for $\mathcal{O}(n)$ this is explicit; for general $\mathcal{L}$ this follows from the construction in Definition 6). The divisor $D' := \mathrm{div}(s) + n[\infty]$ satisfies $\mathcal{O}(D') \cong \mathcal{L}$. By the key lemma, $D' \sim n[\infty]$, so by Theorem 5, $\mathcal{L} \cong \mathcal{O}(D') \cong \mathcal{O}(n[\infty]) = \mathcal{O}(n)$. Hence every degree-$n$ line bundle is isomorphic to $\mathcal{O}(n)$, so degree is a complete invariant.

Why no circular dependency: The surjectivity of Div $\to$ Pic (every line bundle comes from a divisor) also follows from the key lemma: $\mathcal{O}(n) = \mathcal{O}(n[\infty])$ comes from $n[\infty]$. This is independent of the injectivity argument; the key lemma is the sole input. □

Why $\mathbb{CP}^1$ is the simplest case. On a genus-g curve, the degree map $\deg : \mathrm{Pic}(X) \to \mathbb{Z}$ exists, but its kernel $\mathrm{Pic}^0(X)$ (degree-0 line bundles) is a g-dimensional complex torus (the Jacobian). For $\mathbb{CP}^1$ ($g = 0$), Theorem 8 says $\mathrm{Pic}^0(\mathbb{CP}^1) = 0$: every degree-0 bundle is trivial because $D \sim 0$ whenever $\deg D = 0$. This is the sense in which $\mathbb{CP}^1$ is the simplest compact Riemann surface.



$\mathrm{Pic}(\mathbb{CP}^1) \cong \mathbb{Z}$: winding number is the only invariant

3.4 The Short Exact Sequence

Remark 7. By Liouville's theorem, every holomorphic function on the compact surface $\mathbb{CP}^1$ is constant, so $\mathcal{O}(\mathbb{CP}^1)^\times = \mathbb{C}^\times$. There are no nontrivial globally holomorphic gauge transformations on $\mathbb{CP}^1$.

Proposition 9. *There is a short exact sequence*

$$0 \to \mathbb{C}^\times \to \mathcal{M}(\mathbb{CP}^1)^\times \xrightarrow{\mathrm{div}} \mathrm{Div}(\mathbb{CP}^1) \to \mathrm{Pic}(\mathbb{CP}^1) \to 0.$$

Proof (sketch). **Exactness at $\mathcal{M}^\times$:** A function with all winding numbers $\mathrm{ord}_p(f) = 0$ has $\mathrm{div}(f) = 0$, hence is holomorphic and nonvanishing on compact $\mathbb{CP}^1$, hence constant.

Exactness at Div**:** By definition, image(div) = Prin($\mathbb{CP}^1$) = ker(Div $\to$ Pic).

Exactness at Pic**:** Every $\mathcal{O}(n) = \mathcal{O}(n[\infty])$ comes from the divisor $n[\infty]$, so Div $\to$ Pic is surjective. □

Exactness at $\mathrm{Pic}(\mathbb{CP}^1)$ — every line bundle arises from a divisor — requires $H^1(\mathbb{CP}^1, \mathcal{M}^\times) = 0$. This follows from Theorem 8 (every bundle is $\mathcal{O}(n[\infty])$), not from any formal property of the exact sequence alone.

4 The Canonical Bundle and Serre Duality

The canonical bundle is determined by how 1-forms transform on the overlap — another winding number calculation.

4.1 The Canonical Bundle $K_{\mathbb{CP}^1}$

Canonical Bundle

Definition 9. The **canonical bundle** $K_X = \Omega^1_X$ is the cotangent line bundle; its sections over U are holomorphic 1-forms on U.

Remark 8 (Transition function via the winding form)**.** On $U_0 \cap U_\infty$, a 1-form $f(z)\,dz$ in the z-trivialization transforms via $w = 1/z$:

$$f(z)\,dz = f\left(\tfrac{1}{w}\right)\cdot d\left(\tfrac{1}{w}\right) = f\left(\tfrac{1}{w}\right)\cdot\left(-\frac{1}{w^2}\right)dw = -\frac{f(1/w)}{w^2}\,dw.$$

The section in the w-trivialization is $s_\infty(w) = -f(1/w)/w^2$. The gluing $s_0(z) = g^K_{0\infty}(z)\cdot s_\infty(1/z)$ gives $f(z) = g^K_{0\infty}(z)\cdot(-z^2 f(z))$, hence

$$g^K_{0\infty}(z) = -z^{-2}.$$

The winding number: since $d(-z^{-2}) = 2z^{-3}\,dz$,

$$\deg K = \frac{1}{2\pi i}\oint_{|z|=1}\frac{d(-z^{-2})}{-z^{-2}} = \frac{1}{2\pi i}\oint_{|z|=1}\frac{2z^{-3}\,dz}{-z^{-2}} = \frac{1}{2\pi i}\oint_{|z|=1}\frac{-2\,dz}{z} = -2.$$

Therefore $K_{\mathbb{CP}^1} = \mathcal{O}(-2)$.

Observation (Consistency check via dz/z)**.** In the z-chart, dz/z has a simple pole at 0. In the w-chart ($w = 1/z$): $\dfrac{dz}{z} = \dfrac{-w^{-2}\,dw}{w^{-1}} = -\dfrac{dw}{w}$, a simple pole at $w = 0$ (i.e., at ∞). So

$$\operatorname{div}\left(\frac{dz}{z}\right) = -[0] - [\infty], \qquad \deg = -2 = \deg K. \checkmark$$

All meromorphic 1-forms on $\mathbb{CP}^1$ have divisor of degree -2; none are holomorphic everywhere, consistent with $h^0(K) = \ell(K) = \max(0, -2+1) = 0$.

Remark 9. $\deg K = -2$ is consistent with $g = 0$ and the **genus formula** $\deg K = 2g - 2$ (a consequence of Gauss–Bonnet applied to the curvature of K_X). Riemann–Hurwitz, which governs ramification of covering maps, is the wrong reference here.

4.2 Serre Duality

Theorem 10 (Serre Duality — statement only)**.** *For a holomorphic line bundle $\mathcal{L}$ on a compact Riemann surface X,*

$$H^1(X, \mathcal{L}) \cong H^0\big(X,\, K_X \otimes \mathcal{L}^{-1}\big)^\vee$$

as $\mathbb{C}$-vector spaces, where $(-)^\vee = \mathrm{Hom}_{\mathbb{C}}(-, \mathbb{C})$.

The proof of Serre duality requires fine sheaves, the $\bar{\partial}$-operator, and elliptic regularity. It is deferred.

Remark 10 (Consequence for $\mathbb{CP}^1$)**.** With $\mathcal{L} = \mathcal{O}(n)$ and $K_{\mathbb{CP}^1} = \mathcal{O}(-2)$:

$$H^1\big(\mathbb{CP}^1,\, \mathcal{O}(n)\big) \cong H^0\big(\mathbb{CP}^1,\, \mathcal{O}(-n-2)\big)^\vee.$$

Taking dimensions (the dual has the same dimension):

$$h^1\big(\mathcal{O}(n)\big) = h^0\big(\mathcal{O}(-n-2)\big) = \max(0,\, -n-1).$$

5 Riemann–Roch Theorem on $\mathbb{CP}^1$

5.1 Statement

On a compact Riemann surface X of genus g:

$$\ell(D) - \ell(K - D) = \deg D + 1 - g.$$

For $X = \mathbb{CP}^1$ ($g = 0$):

$$\ell(D) - \ell(K - D) = \deg D + 1.$$

Viewed as a winding-number balance:

$$\underbrace{\ell(D)}_{\text{sections: } \mathrm{ord}_p \geq -n_p} - \underbrace{\ell(K - D)}_{h^1 \text{ via Serre duality}} = \underbrace{\deg D}_{\text{total winding}} + 1.$$

Theorem 11 (Riemann–Roch on $\mathbb{CP}^1$)**.** *For every $D \in \mathrm{Div}(\mathbb{CP}^1)$,*

$$\ell(D) - \ell(K - D) = \deg D + 1.$$

5.2 Proof via Winding Numbers

Proof. **Step 1: Reduce to $D = n[\infty]$.** Every finite point satisfies $[a] \sim [\infty]$ via $\mathrm{div}(z - a) = [a] - [\infty]$ (the winding form $d(z - a)/(z - a)$ has winding $+1$ at a and -1 at ∞). Applying repeatedly, every D of degree n satisfies $D \sim n[\infty]$. By Corollary 6: $\ell(D) = \ell(n[\infty])$ and $\ell(K - D) = \ell(K - n[\infty])$.

Step 2: Compute $\ell(n[\infty])$. By Proposition 2, $\ell(n[\infty]) = \max(0, n + 1)$.

Step 3: Identify the canonical divisor. $K_{\mathbb{CP}^1} = \mathcal{O}(-2)$ has degree -2 (§4.1). By Theorem 8 ($\mathrm{Pic}^0(\mathbb{CP}^1) = 0$), the unique degree-$(-2)$ bundle is $\mathcal{O}(-2) = \mathcal{O}(-2[\infty])$, so take $K = -2[\infty]$. Then $K - D \sim (-n - 2)[\infty]$, and $\ell(K - D) = \max(0, -n - 1)$.

Step 4: Verify case by case.

(i) $n \geq 0$: $\ell(D) = n + 1$, $\ell(K - D) = 0$ (since $-n - 1 < 0$). $(n + 1) - 0 = n + 1$. ✓

(ii) $n = -1$: $\ell(D) = \max(0, 0) = 0$, $\ell(K - D) = \ell(-[\infty]) = \max(0, 0) = 0$. $0 - 0 = 0 = -1 + 1$. ✓

(iii) $n \leq -2$: $\ell(D) = 0$, $\ell(K - D) = \max(0, -n - 1) = -n - 1 > 0$. $0 - (-n - 1) = n + 1$. ✓

□

Riemann–Roch on $\mathbb{CP}^1$ is a bookkeeping identity for winding numbers: the number of independent sections of $\mathcal{O}(n)$ minus the number of independent correction sections equals the total winding n plus one.

5.3 Cohomological Reformulation

Proposition 12. *Riemann–Roch on $\mathbb{CP}^1$ is equivalent to*

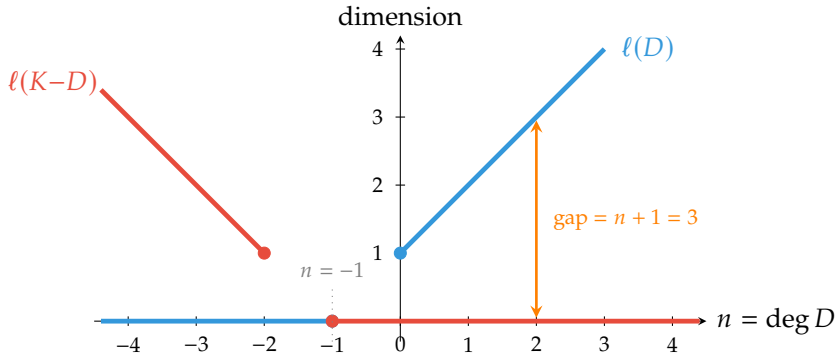
$$h^0\big(\mathcal{O}(D)\big) - h^1\big(\mathcal{O}(D)\big) = \deg D + 1.$$

Proof. $h^0(\mathcal{O}(D)) = \ell(D)$ by Theorem 4. By Serre duality (Theorem 10) the vector-space isomorphism $H^1(\mathbb{CP}^1, \mathcal{O}(n)) \cong H^0(\mathbb{CP}^1, \mathcal{O}(-n-2))^\vee$ gives the dimension equality $h^1(\mathcal{O}(n)) = h^0(\mathcal{O}(-n-2)) = \ell(K-D)$ (Remark 10). Substituting into Theorem 11 yields the result. □

Observation. Three distinct regimes:

- $n \geq 0$: $\ell(D) = n + 1 > 0$ and $h^1 = 0$. Riemann–Roch is the clean count $h^0 = \deg D + 1$.
- $n = -1$: $\ell(D) = 0$ and $h^1 = 0$. Both sides are 0; Riemann–Roch holds trivially.
- $n \leq -2$: $\ell(D) = 0$ and $h^1 = -n - 1 > 0$. Serre duality "reflects" the count: $0 - (-n - 1) = n + 1$.

Causal order: $\ell(D) = 0$ already at $n = -1$, independently of h^1. The h^1 obstruction activates only at $n \leq -2$; it does not **cause** $\ell(D)$ to vanish, but accounts for the deficit $\deg D + 1 < 0$ once $\ell(D)$ is already zero.



$$\ell(D) - \ell(K - D) = n + 1 = \deg D + 1 \quad \text{for all } n \in \mathbb{Z}$$

6 Examples and Applications

6.1 Table of $\mathcal{O}(n)$ Sections

n	Bundle	$\ell(n[\infty])$	Basis of $H^0(\mathcal{O}(n))$
$n \le -1$	$\mathcal{O}(n)$	0	$\emptyset$
-2	$K_{\mathbb{CP}^1} = \mathcal{O}(-2)$	0	$\emptyset$
-1	$\mathcal{O}(-1)$	0	$\emptyset$
0	$\mathcal{O}_{\mathbb{CP}^1}$	1	$\{1\}$
1	$\mathcal{O}(1)$	2	$\{1, z\}$
2	$\mathcal{O}(2)$	3	$\{1, z, z^2\}$
$n \ge 0$	$\mathcal{O}(n)$	$n+1$	$\{1, z, \ldots, z^n\}$

6.2 Verifying Riemann–Roch on Specific Divisors

The following three examples revisit computations from [1] and restate them in terms of line bundles. Throughout, the affine point $z_0 \in \mathbb{C}$ corresponds to $[1 : z_0] \in \mathbb{CP}^1$.

Example 3 ($D = [0]$). $\deg D = 1$, so $\mathcal{O}([0]) \cong \mathcal{O}(1)$. By Proposition 2: $\ell = 1 + 1 = 2$, basis $\{1, z\}$. Riemann–Roch: $2 - 0 = 1 + 1$. ✓

Example 4 ($D = [0] + [1]$). $\deg D = 2$, so $\mathcal{O}([0] + [1]) \cong \mathcal{O}(2)$. $\ell = 2 + 1 = 3$, basis $\{1, z, z^2\}$. Riemann–Roch: $3 - 0 = 2 + 1$. ✓

Example 5 ($D = [\infty] - [1]$). $\deg D = 0$, so $\mathcal{O}([\infty] - [1]) \cong \mathcal{O}(0) = \mathcal{O}$. $\ell = \max(0, 1) = 1$, basis $\{1\}$ (constants). Cross-check from [1]: $L([\infty] - [1]) = \mathrm{span}_{\mathbb{C}}\{z - 1\}$, dimension 1. ✓Riemann–Roch: $1 - 0 = 0 + 1$. ✓

6.3 Morphisms to Projective Space

Proposition 13. *If $\deg D \ge 1$ on $\mathbb{CP}^1$, a basis $\{f_0, \ldots, f_m\}$ of $L(D)$ ($m = \ell(D) - 1$) defines a morphism*

$$\phi_D : \mathbb{CP}^1 \to \mathbb{P}^m, \qquad p \mapsto [f_0(p) : \cdots : f_m(p)].$$

The base locus is empty since the constant function $1 \in L(D)$ never vanishes.

Convention: the affine point $z \in \mathbb{C}$ corresponds to the homogeneous point $[1 : z] \in \mathbb{CP}^1$.

Example 6 ($D = [\infty]$, identity map)**.** Basis $\{1, z\}$, target $\mathbb{P}^1$.

$$\phi : [1 : z_0] \mapsto [1 : z_0].$$

This is the identity $\mathbb{CP}^1 \xrightarrow{\sim} \mathbb{CP}^1$.

Example 7 ($D = 2[\infty]$, Veronese embedding)**.** Basis $\{1, z, z^2\}$, target $\mathbb{P}^2$.

$$\phi : [1 : z] \mapsto [1 : z : z^2].$$

The image is the conic $\{[X : Y : Z] \in \mathbb{P}^2 : XZ = Y^2\}$.

References

[1] Ji, Yonghyeon. *Riemann–Roch Theorem on* $\mathbb{CP}^1$. Lecture notes, 2025.